

Le Hoang Tran

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SUMMARY

Backend-focused Software Engineer with strong experience in building large-scale systems. Specialized in high-concurrency, low-latency, cloud-native infrastructure and performance optimization. Delivered products that serve millions of users.

EDUCATION

George Mason University

M.S. in Computer Science | GPA: 3.86

(Expected) Dec 2026

Fairfax, VA, USA

Hanoi University of Science and Technology

B.S. in Information Technology | GPA: 3.62 (top 5% of cohort)

Aug 2017 – May 2021

Hanoi, Vietnam

EXPERIENCE

Viettel Digital

Backend Software Engineer

May 2021 – Jun 2025

Hanoi, Vietnam

Customer Engagement Platform

2023 – 2025

- Designed and implemented an **in-house marketing platform** supporting personalized campaigns through push notifications, in-app messaging, and rewards distribution, processing **100M+ events** daily.
- Increased user segmentation creation throughput from **500 to 35,000 QPS** while **reducing p95 latency by 80%** by introducing **Roaring Bitmap-based solution** and multi-layer caching.
- Engineered **workflow automation module** using Camunda for business process orchestration and rule evaluation, enabling non-technical operations teams to configure and manage marketing workflows independently.
- Led a team of 4 backend engineers** through sprint planning, system design, and code reviews, continuously delivering new features to meet evolving product requirements.
- Owned end-to-end system observability using **Grafana**, identifying performance bottlenecks and improving reliability through configuration tuning, caching strategies, and microservices refactoring.

Data Tracking System

2022 – 2023

- Designed and delivered a high-throughput **in-house event tracking system** for mobile and web platforms, processing **150M+ events daily** and replacing a third-party tracking provider to **reduce licensing costs**.
- Implemented an online tracking rule configuration module supporting 3,000+ dynamic rules, reducing event validation latency by **30%** by replacing linear rule processing with a decision tree-based validation engine.
- Built non-blocking ingestion APIs using **Spring WebFlux**, supporting **1700+ RPS** on average for high-volume event ingestion while reducing request-per-thread overhead.

Smart Authentication System

2021 – 2022

- Collaborated on the research and delivery of **SmartOTP** solution for **20M+ end users**, eliminating reliance on SMS OTPs, **saving over \$200K annually** and enhancing authentication reliability.
- Implemented secure key exchange, OTP generation, and digital signature mechanisms across critical APIs, strengthening authentication integrity and reducing unauthorized access risks.
- Hardened Android application security using native **C/JNI**, applying obfuscation and root/jailbreak/tamper detection, successfully passing enterprise-level security audits.

User Contact System

2024

- Designed and implemented a backend module to collect and synchronize user contact data across multiple devices for the Viettel Money application, supporting **20M+ end users**.
- Optimized contact storage with Roaring Bitmap, reducing storage usage by **70%** and lowering user matching latency.

TECHNICAL SKILLS

Languages: Java, Python, Go, C

Backend: Spring Framework, Spring Boot, Camunda, REST, gRPC, JUnit, Mockito, Testcontainers

Infra & Data: Apache Kafka, Redis, MySQL, MongoDB, Elasticsearch, MinIO, Clickhouse

Platform & Ops: Kubernetes, Docker, Jenkins, Grafana, ELK, K6, Linux, Bash, Git

AI Tooling: Claude, OpenAI Codex, Agentic Workflows, Prompt Engineering

CERTIFICATES

AWS Certified Solutions Architect – Associate

Sep 2023

Oracle Certified Java SE Programmer

Apr 2023